Curriculum Vitae

Yan Zhou, Ph.D.
Plant Science Researcher (Research Scientist III)
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Education

2015 - 2022	Ph.D. in Genetics and Genomics, Iowa State University (Advisor: Dr. Patrick S. Schnable)
2009 - 2012	M.S. in Genetics, Nanjing Agriculture University, China (Advisor: Dr. Zheng-Qiang Ma)
2005 - 2009	B.S. in Plant Science, Agricultural University of Hebei, China

Employment

Mar. 2024 - Present	Plant Science Researcher (Research Scientist III)
	Iowa State University
Aug. 2022 - Feb. 2024	Postdoctoral Research Associate
	Iowa State University
Aug. 2015 - July 2022	Graduate Research Assistant
	Iowa State University
Apr. 2014 - Jan. 2015	Research Associate
	Beijing Shou jia Li Hua Bio-tech Co. Ltd., China
Apr. 2013 - Mar. 2014	Research Associate
	Beijing Golden Guanfeng Biotech Ltd., China

Research Interests

-Multi-omic characterization and functional analysis of genes regulating light-dependent leaf azimuthal re-orientation.

Goals: 1) Characterizing canopy azimuthal orientations in a maize diversity population under high planting density using UAV and digital imaging platforms; 2) Evaluating the impacts of canopy azimuthal orientations on light interception; 3) Identifying genes that are critical for light capture of the canopy under high planting density and shading conditions in maize.

Roles: 1) Developing pipelines for high-throughput phenotyping of light interception rate and data analyses; 2) Detecting loci contribute to the phenotypic variation of traits of interest using genome-wide association studies (GWAS); 3) Conducting transcriptional analyses on the identified candidate genes using *Mu*-insertion mutants to explore the gene-regulation pathways that participate density and shading response

in maize; 4) Studying canopy and root developments of the mutants under light/shade and other abiotic stress.

-Investigating the impacts of breeding selections on tolerance to high planting density and abiotic stress in modern maize germplasm.

Goals: 1) Testing the hypothesis that breeding selections on grain yield has indirectly changed the morphological response to high density planting in maize using inbred lines released in two breeding eras (before 1951 vs. after 1980); 2) Identifying genomic regions carry alleles that are beneficial to the tolerance of abiotic stresses and high density planting in the germplasm from early breeding era that have not been utilized in moder germplasm.

Roles: 1) Conducting analyses to evaluate the changes in density response of the collected morphological traits between two breeding eras; 2) Evaluating the tolerance to abiotic stress of germplasm from two breeding eras; 3) Conducting genomic screens to identify haplotype blocks that are potentially beneficial to light capture, regulating photosynthesis, and energy management.

-Predictive modeling of the density-dependent morphological responses and their impact on grain yield in maize.

Goals: 1) Investigating the roles of light-sensing genes in density-dependent morphological responses; 2) Identifying signaling pathways in leaf and stalk that are critical for response to high planting density; 3) Improving the photosynthesis efficiency and grain yield under high planting density.

Roles: 1) Using multi-omic approaches to investigate signaling pathways in leaf and stalk that are critical for response to high planting density. 2) Generating gene-edited mutants in cryptochrome, phytochrome, and phototropin genes; 3) Building and testing prediction models to link plant architecture traits to high light capture and grain yield.

Peer-Reviewed Publications (13)

1. Saleem, N., Jubery, T.Z., Balu, A., **Zhou, Y.**, Li, Y., Schnable, P.S., Krishnamurthy, A., and Ganapathysubramanian, B., 2025. Accessing the Effect of Phyllotaxy and Planting Density on Light Use Efficiency in Field-Grown Maize using 3D Reconstructions. *Smart Agricultural Technology, pp.101566.*
2. **Zhou, Y.,** Kusmec, A., Schnable, P.S., 2024. Genetic regulation of self-organizing azimuthal canopy orientations and their impacts on light interception in maize. *The Plant Cell*, *36(5), pp.1600-1621*.
3. Hou, Q., An, X., Ma, B., Wu, S., Wei, X., Yan, T., **Zhou, Y.**, Zhu, T., Xie, K., Zhang, D., and Li, Z., et al., 2023. ZmMS1/ZmLBD30-orchestrated transcriptional regulatory networks precisely control pollen exine development. *Molecular Plant*, *16(8), pp.1321-1388.*
4. **Zhou, Y.,** Kusmec, A., Mirnezami, S.V., Attigala, L., Srinivasan, S., Jubery, T.Z., Schnable, J.C., Salas-Fernandez, M.G., Ganapathysubramanian, B., and Schnable, P.S., 2021. Identification and utilization of genetic determinants of

trait measurement errors in image-based, high-throughput phenotyping. *The Plant Cell*, *33(8), pp.2562-2582*.

5. Mirnezami, S.V., Srinivasan, S., **Zhou, Y.,** Schnable, P.S., and Ganapathysubramanian, B., 2021. Detection of the progression of anthesis in fieldgrown maize tassels: A case study. *Plant Phenomics*, *2021- 4238701*.
6. Li, M., Cao, L., Mwimba, M., **Zhou, Y.,** Li, L., Zhou, M., Schnable, P.S., O'Rourke, J.A., Dong, X., and Wang, W., 2019. Comprehensive mapping of abiotic stress inputs into the soybean circadian clock. *Proceedings of the National Academy of Sciences*, *116(47), pp.23840-23849*.
7. **Zhou, Y.,** Srinivasan, S., Mirnezami, S.V., Kusmec, A., Fu, Q., Attigala, L., Salas Fernandez, M.G., Ganapathysubramanian, B., and Schnable, P.S., 2019. Semiautomated feature extraction from RGB images for sorghum panicle architecture GWAS. *Plant physiology, 179(1), pp.24-37*.
8. An, X., Dong, Z., Tian, Y., Xie, K., Wu, S., Zhu, T., Zhang, D., **Zhou, Y.**, Niu, C., Ma, B., Hou, Q., et al., 2019. *ZmMs30* encoding a novel GDSL lipase is essential for male fertility and valuable for hybrid breeding in maize. *Molecular Plant*, *12(3), pp.343-359*.
9. Xie, K., Wu, S., Li, Z., **Zhou, Y.**, Zhang, D., Dong, Z., An, X., Zhu, T., Zhang, S., Liu, S., Li, J., Wang, X., 2018. Map-based cloning and characterization of Zea mays *male sterility33* (*ZmMs33*) gene, encoding a glycerol-3-phosphate acyltransferase. *Theoretical and Applied Genetics*, *131, pp.1363-1378*.
10. Zhang, D., Wu, S., An, X., Xie, K., Dong, Z., **Zhou, Y.,** Xu, L., Fang, W., Liu, S., Liu, S., and Zhu, T., et al., 2018. Construction of a multicontrol sterility system for a maize male sterile line and hybrid seed production based on the ZmMs7 gene encoding a PHD-finger transcription factor. *Plant Biotechnology Journal*, *16(2), pp.459-471*.
11. Cao, W., Fu, B., Wu, K., Li, N., **Zhou, Y.**, Gao, Z., Lin, M., Li, G., Wu, X., Ma, Z. and Jia, H., 2014. Construction and characterization of three wheat bacterial artificial chromosome libraries. *International Journal of Molecular Sciences*, *15(12), pp.21896-21912*.
12. Xue, S., Xu, F., Li, G., **Zhou, Y**, Lin, M., Gao, Z., Su, X., Xu ,X., Jiang, G., Zhang, S., Jia, H., Kong, Z., Ma, Z., 2013. Fine mapping *TaFLW1*, a major QTL controlling flag leaf width in bread wheat (Triticum aestivum L.) *Theoretical and applied genetics, 126(8), pp.1941-1949.*
13. Xue, S., Xu, F., Tang, M., **Zhou, Y.,** Li, G., An, X., Lin, F., Xu, H., Jia, H., Zhang, L. Kong, Z., 2011, Ma, Z., 2011. Precise mapping *Fhb5*, a major QTL conditioning resistance to Fusarium infection in bread wheat (Triticum aestivum L.). *Theoretical and applied genetics*, *123(6), pp.1055-1063.*

Manuscripts in Preparation and Pre-prints

1. **Zhou,Y.** Hickman, J., Jubery, Z.T., Ganapathysubramanian, B., Schnable, P. S. (*in preparation*) Unravelling The Pleiotropic Effects of Genes on Canopy, Below-ground Architecture and Response to Increased Planting Density in Maize.

2. **Zhou. Y.,** Kusmec A., Coffey L., Yeh C.-T. “Eddy”, and Schnable P.S. (*in preparation*) Analysis of A Public Maize ERA Inbred Panel Reveals Changes in Density-dependent Morphological Responses Over 50 Years of Breeding Selection.
3. Saleem, N., Jubery, T.Z., **Zhou, Y.**, Li, Y., Krishnamurthy, A., Schnable, P.S., and Ganapathysubramanian, B., 2025. Towards smart canopies: Algorithmic design of maize canopy architectures that maximize light use efficiency. *arXiv preprint arXiv:2512.06064*.
4. Zeng, Q., Cao, W., Li, W., Wu, J., Figueroa, M., Liu, H., Qin, G., Wang, Q., Yang, L., **Zhou, Y.** and Yu, Y., et al., 2022. Near telomere-to-telomere nuclear phased chromosomes of the dikaryotic wheat fungus Rhizoctonia cerealis. *bioRxiv*, *pp.2022-03*.
5. Zeng, Q., Cao, W., Xing, L., Qin, G., Wu, J., Nagle, M.F., Xiong, Q., Chen, J., Yang, L., Bajaj, P. and Chitikineni, A., **Zhou, Y.**, et al., 2020. A novel high-accuracy genome assembly method utilizing a high-throughput workflow. *BioRxiv*, *pp.2020-11*.

Patents (1)

CN104946660B. Wan X., Wu S., **Zhou Y.**, Xie K., Li J. The DNA sequence and coded protein of the post-meiosis developmental gene *Ms30* in maize. (2017)

Talks (10 = 4 international and 6 national)

Invited

1. Jan 12, 2026. Unravelling pleiotropic effects of genes that regulate canopy and root system architecture and their responses to planting densities in maize. 33rd Plant and Animal Genome Conference, San Diego, CA.
2. Jan 9, 2025. Genetic regulation of self-organizing azimuthal canopy orientations and transcriptome analyses on shade-sensing genes in maize. 32nd Plant and Animal Genome Conference, San Diego, CA.
3. Nov 10, 2023. Genetic regulation of self-organizing azimuthal canopy orientations and their impacts on light interception in maize. Huazhong Agriculture University, Wuhan, Hubei, China.
4. Nov 6, 2023. Genetic regulation of self-organizing azimuthal canopy orientations and their impacts on light interception in maize. Nanjing Agriculture University, Nanjing, Jiangsu, China.
5. Oct 17, 2023. Genetic regulation of self-organizing azimuthal canopy orientations and their impacts on light interception in maize. China Agriculture University, Beijing, China.

Contributed

1. Jan 31, 2024. Genetic regulation of self-organizing azimuthal canopy orientations and their impacts on light interception in maize. CBC seminar series, Iowa State University, Ames, IA.

2. March 16, 2023. Genetic regulation of self-organizing canopy patterns and their impacts on light interception in maize. Corn Breeding Research (CBR) Meeting 2023, St. Louis, MO.
3. March 09, 2021. Identification of genetic determinants of trait measurement errors in image-based, high-throughput phenotyping. 63rd Annual Maize Genetics Meeting, Virtual.
4. Feb 18, 2021. Identification of genetic determinants of trait measurement errors in image-based, high-throughput phenotyping. Corn Breeding Research (CBR) Meeting 2021, Virtual.
5. July 27, 2020. Identification of genetic determinants of phenotyping errors in image-based high-throughput phenotyping. Plant Biology 2020, Virtual.

Awards and Honors

2025	Peter J. Loesch, Jr. Memorial Fund Travel Award, Iowa State University
2024	Editor's Citation for Excellence for Reviewers, The Plant Genome
2022	C.R. Weber Award for Excellence in Planting Breeding, Iowa State University
2020	Bayer Travel Award, Bayer Crop Science
2020	Sui Tong Chan Fung Travel Award, Iowa State University
2019	Francis and Beatriz (Castro) Mar Graduate Scholarship, Iowa State University

Grant Experience

2026 Azimuthal canopy reorientation and its potential contributions to improved maize yields. **USDA-NIFA** (Roles: Co-PI; Ranked as outstanding; Not funded)

Contributed Grants

2023 NIFA AG2PI Collaborative: Improving Causal Gene Detection across Crop and Livestock Species. **USDA-NIFA**: 2023-70412-41087 ($1,132,877).

2017 Integrative analyses of maize shoot patterning across multiple scales. **NSF-PGRP**(not funded)

Teaching and Mentoring

Teaching

2025 Spring	Co-instructor, Advanced Plant Breeding (AGRON 621)
	Topics: Genetics of breeding populations; parent selection and mating design; variance component estimation; QTL mapping; genotype-environment interactions
2025 Spring	Invited Lecturer, Advanced Molecular Genetics (GDCB 511)

	Topics: miRNA and siRNA
2023 Fall	Instructor, Undergraduate Research Credit (BIO 699).
2022 Fall	Instructor, Undergraduate Research Credit (BIO 699).
2018 Fall	Invited Lecturer, Fundamentals of Predictive Plant Phenomics (ME585)
	Topics: Physiology of maize and its field phenotyping

Mentorship

2025 - 2026	Mentor, Iowa State University Honors Program (Trainee: Joel Esparra, Biology major)
2024 - Present	Mentored one MS student and one post-doctoral research associate (Schnable Lab) (Trainees: Abigail Bryan, Dr. Cheng Cheng)
2022 - Present	Trained and supervised two technicians and seven undergraduate lab assistants (Schnable Lab) in field operations and molecular biology lab techniques (Trainees: Jesse Hickman, Abigail Bryan, Kara Hagedorn, Desere' 'L McCargar, Eric Hansen)
2016	Mentor, Iowa State University Honors Program (Trainee: Katherine Lenson, Genetics major)

Selected Poster Presentations as Mentor

*Undergraduate presenter

Zhou, Y., Hansen, E., McCargar, D.*, Hagedorn, K.*, Jubery, Z.T., Ganapathysubramanian, B., Schnable, P., 2025. Genes that Regulate Azimuthal Canopy Re-Orientation Exhibit Sensitivity to Plant Density. 67th Annual Maize Genetics Meeting.

Zhou, Y., Hickman, J., Bryan, A.*, Jubery, Z.T., Ganapathysubramanian, B., Schnable, P., 2024. Analysis of Azimuthal Canopy Re-Orientation Genes. R.F. Baker Plant Breeding Symposium.

Zhou, Y., Hickman, J.*, Schnable, P., 2023. A Time-series Analysis of Azimuthal Canopy Orientation Architecture Using UAV Images. 65th Annual Maize Genetics Meeting.

Leadership and Service

Peer Review Service

Reviewed for Nature Plants, The Plant Journal, Horticulture Research, Plant Phenomics, Journal of Experimental Botany, The Plant Genome, Theoretical and Applied Genetics, Journal of Genetics and Genomics, Molecular Breeding, The Plant Cell Report, Crop Science.

Community Involvement

2024 – Present	Advisor, Iowa State University Kendo Club
2020	Student Organizing Committee, R.F. Baker Plant Breeding Symposium (Corteva Plant Science Symposium Series)

Professional Affiliations

2016 – Present	Member, Maize Genetic Cooperation (MGC)
2020 – Present	Member, American Society of Plant Biologists (ASPB)
2022 – Present	Member, The American Society of Agronomy (ASA), Crop Science Society of America (CSSA), and Soil Science Society of America (SSA)